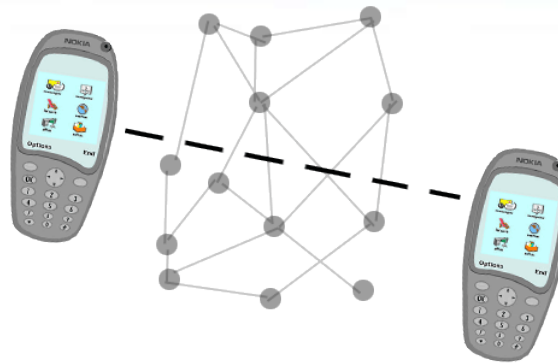




SIP for Mobile Phones

JSR180 - SIP API for J2ME update

JAIN Technology Day, Dallas

26 May 2004

Mikko Lönnfors, Jari Kinnunen, Nokia Research
Center, Helsinki

Contents

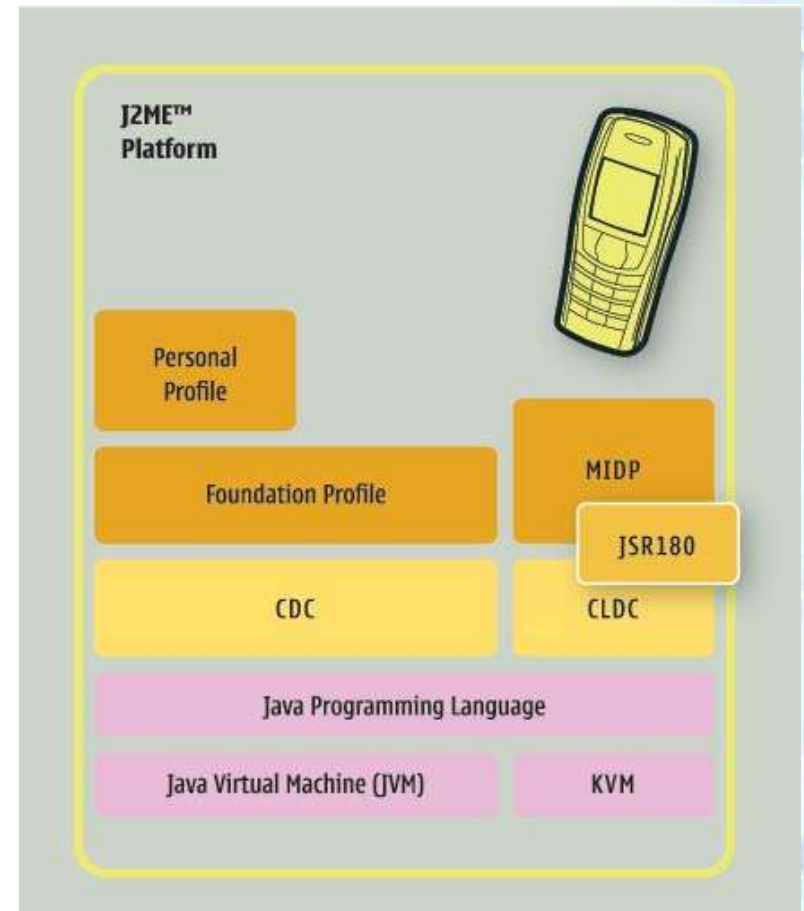
- JSR180 overview
- Goals & Architecture
- Applications
- RI and TCK
- JSR180 status

Standardization Goals

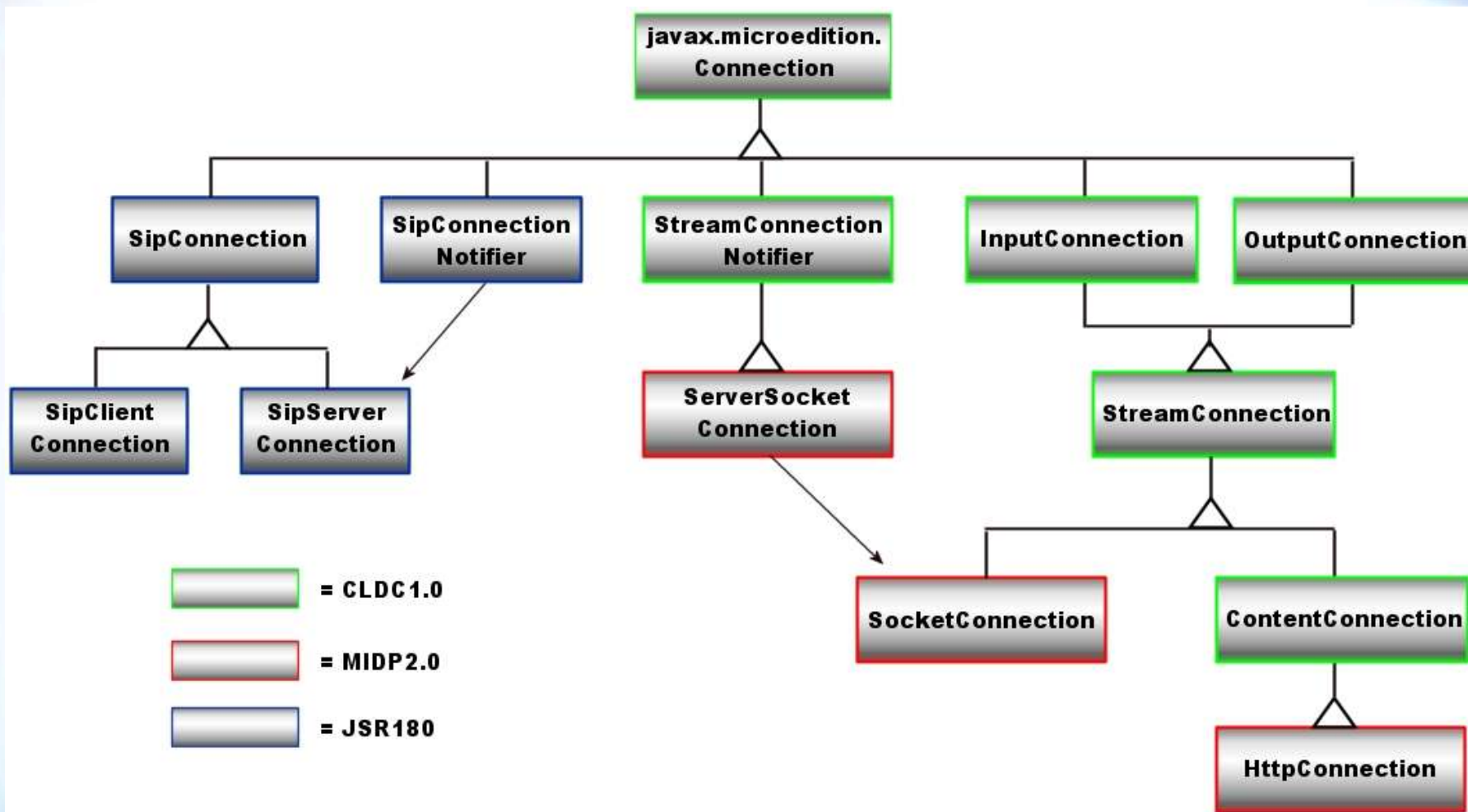
- Enable Rapid SIP Application Development in mobile terminals
- Utilize SIP infrastructure to enable peer-to-peer IP applications
- Integrate to CLDC Generic Connection Framework (GCF)
- Keep the "look and feel" of existing MIDP connectors:
HttpConnection, **SocketConnection**,
MessageConnection, ...
- Simple and lightweight, minimum set of classes, essential functionality
- Provide helper functions for common SIP tasks
- Transaction User level -> flexibility, appl. diversity & freedom
 - compatibility with different SIP flavors (IETF & 3GPP)
- Base on RFC3261. Supports extensions: RFC3265, SIMPLE WG, ...

JSR180 Overview

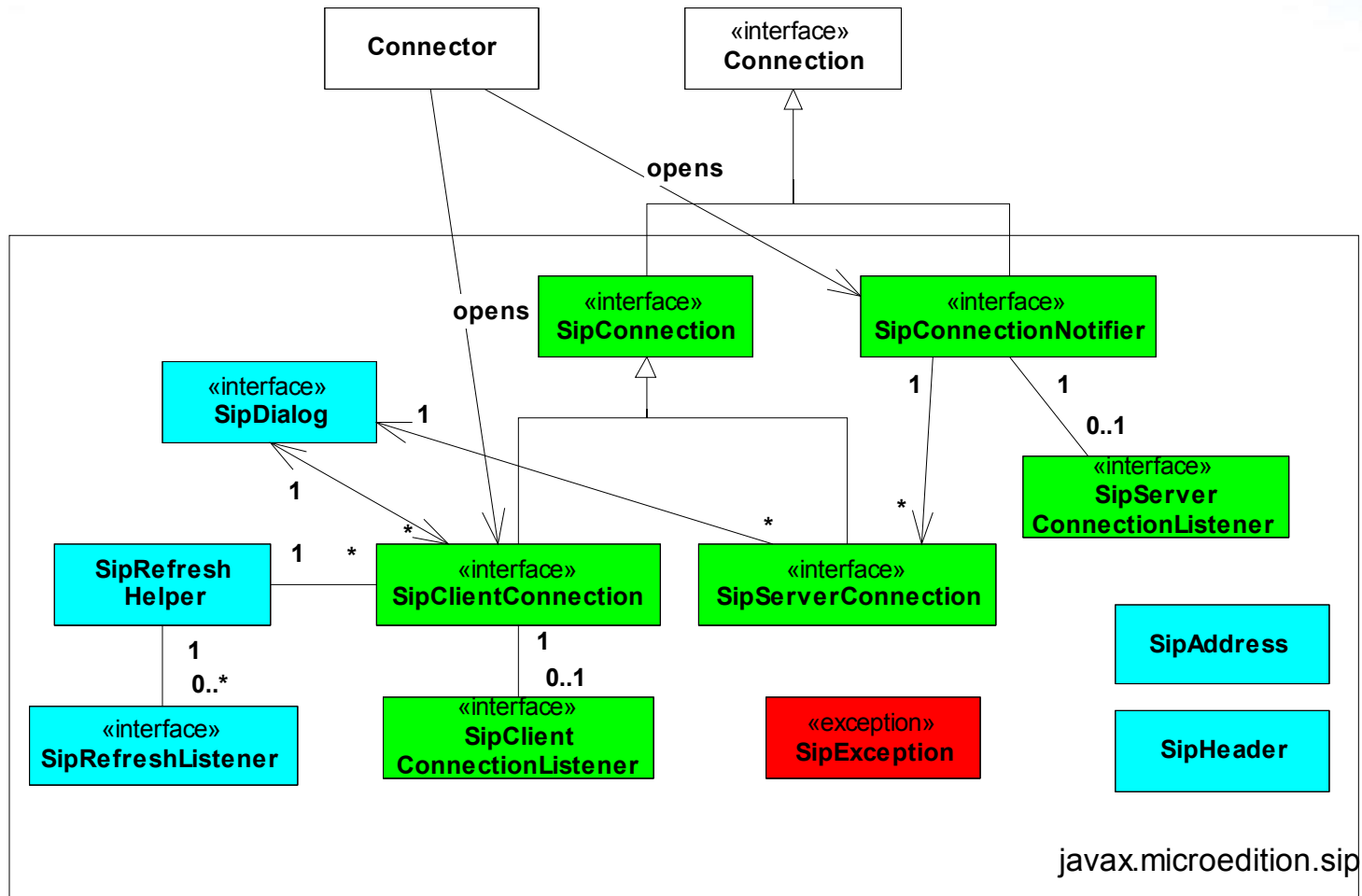
- Standardized Java interface enabling SIP communications in J2ME devices
- Optional Package for J2ME platform, MIDP being the primary target profile
- The specification is based on CLDC1.0 but can be also used with CDC
- Gives Transaction User level access to terminal SIP services
- Support for straightforward and comfortable SIP programming
- Flexible, general level SIP API, which abstract the core network (IETF SIP or 3GPP IMS SIP)



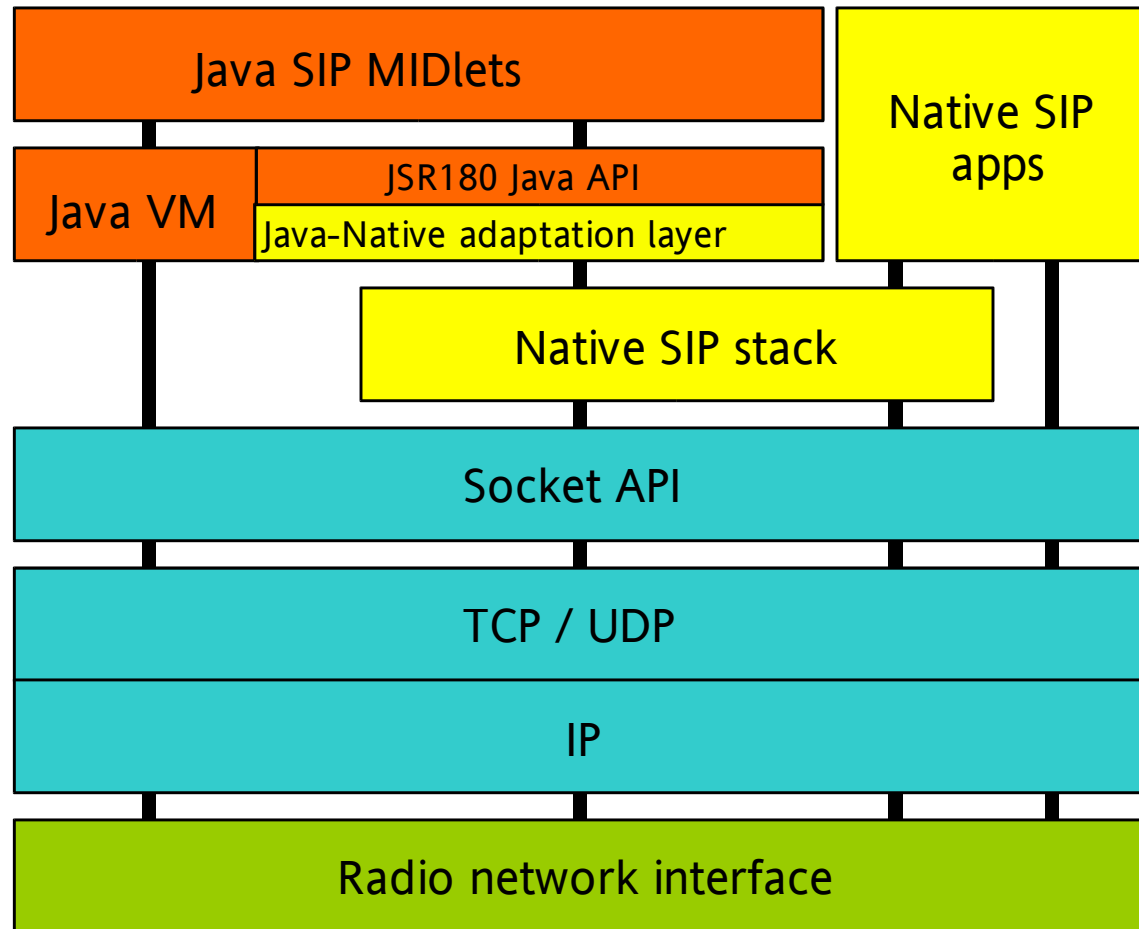
Architecture (SIP in GCF)



Architecture (Class Diagram)



JSR180 API in MIDP terminal



Architecture (UAC <-> UAS)

```
SipClientConnection scc=  
(SipClientConnection)  
Connector.open("sip:userB@host.com");
```

**Terminal A
UAC**

**SipClient
Connection**

```
SipConnectionNotifier scn=  
(SipConnectionNotifier)  
Connector.open("sip:5060");
```

**Terminal B
UAS**

**SipConnection
Notifier**

**SipServer
Connection**

— send
request —>

— receive
request —>

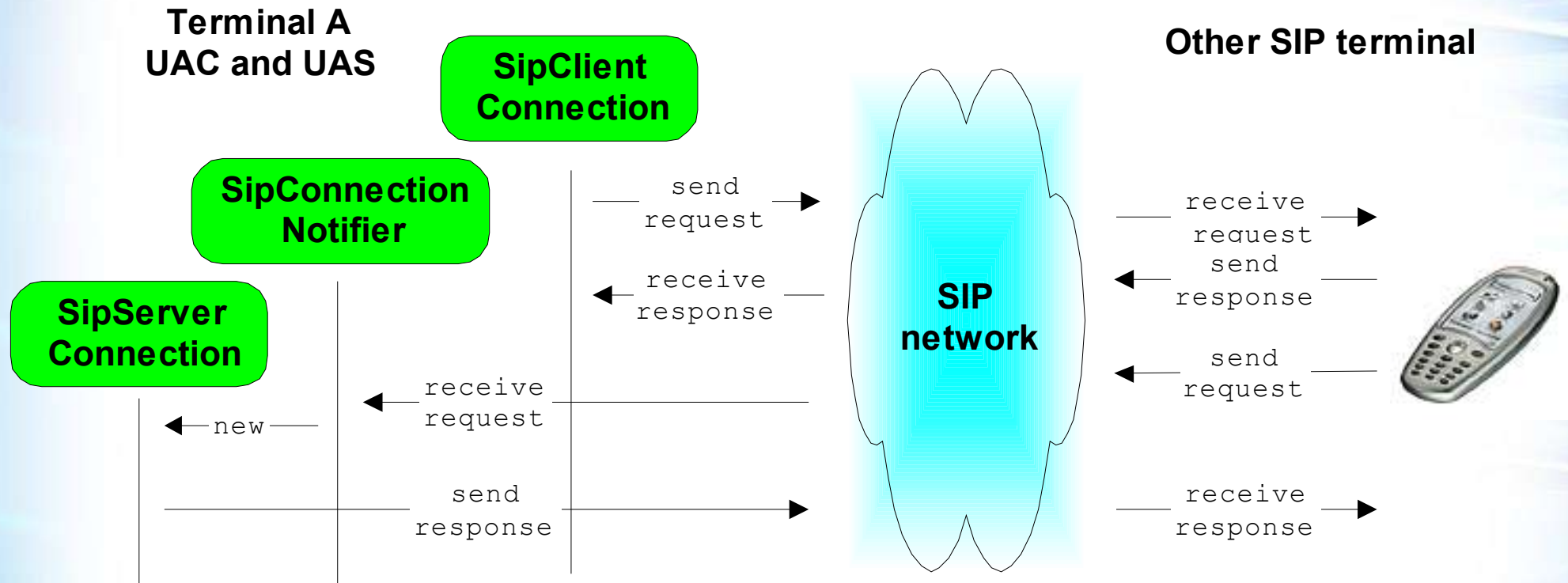
— new —>

← receive
response —

← send
response —

**SIP
network**

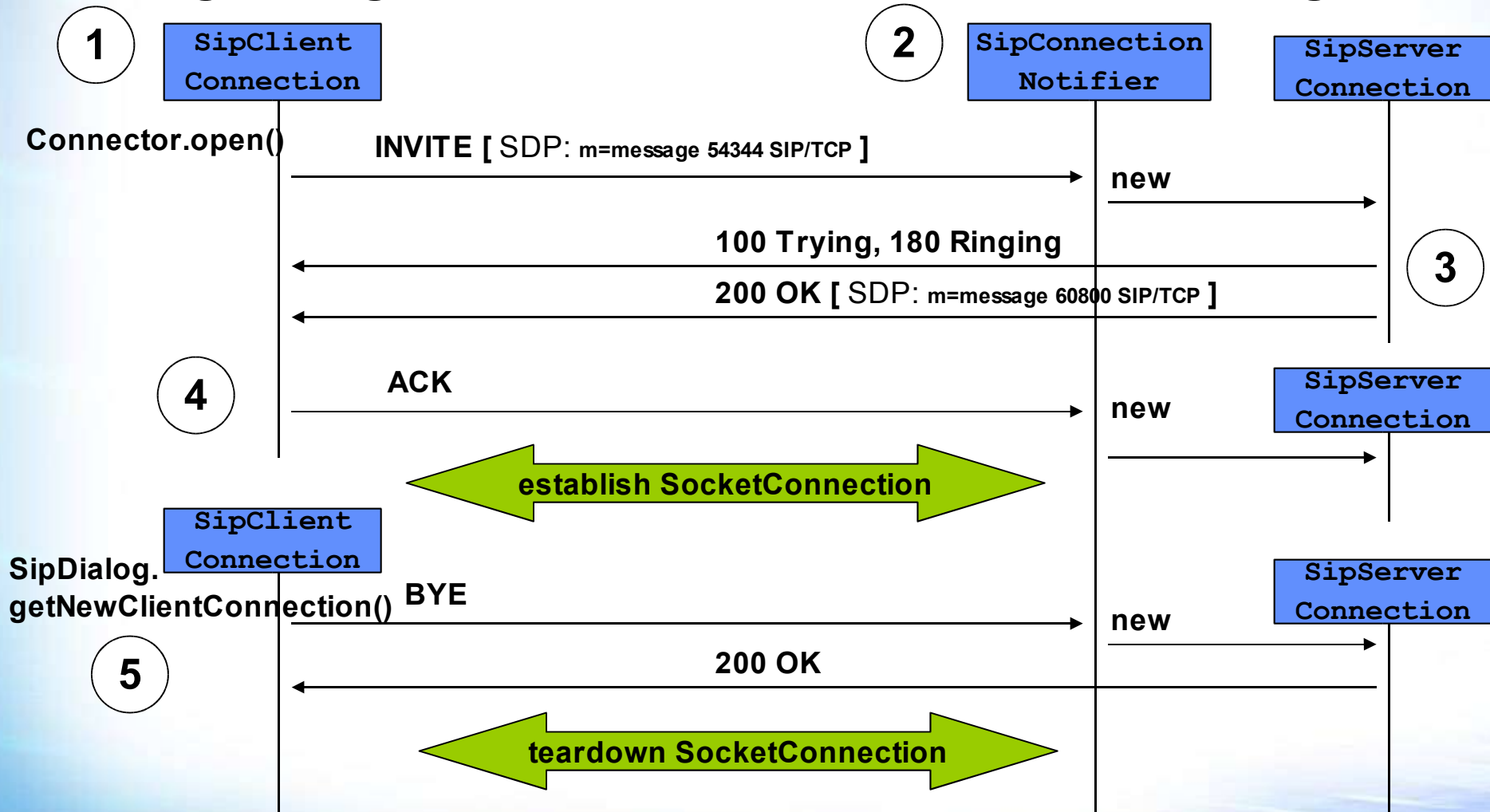
Architecture (UAC & UAS <-> other)



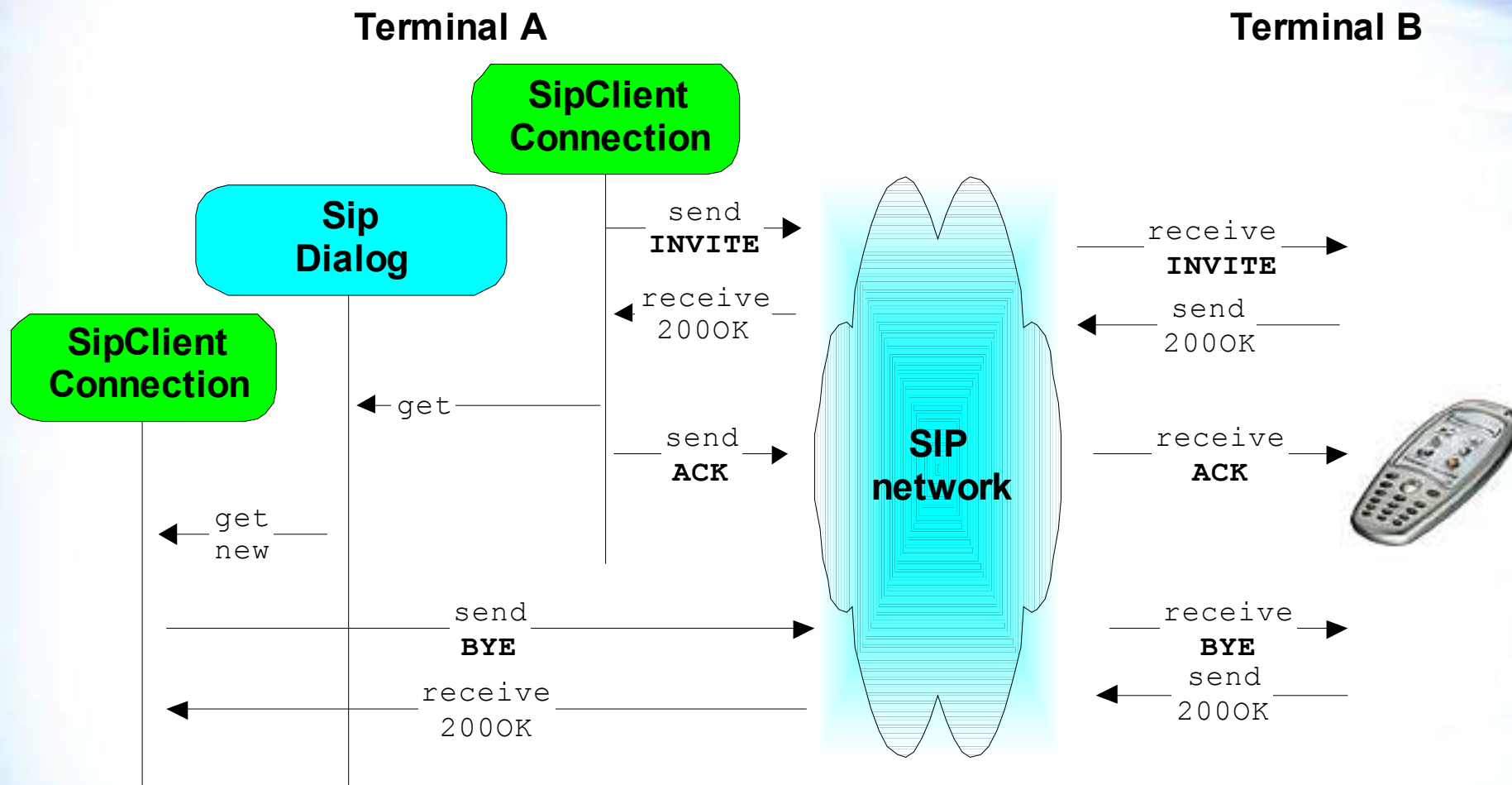
Generic session setup/teardown example

Originating INVITE

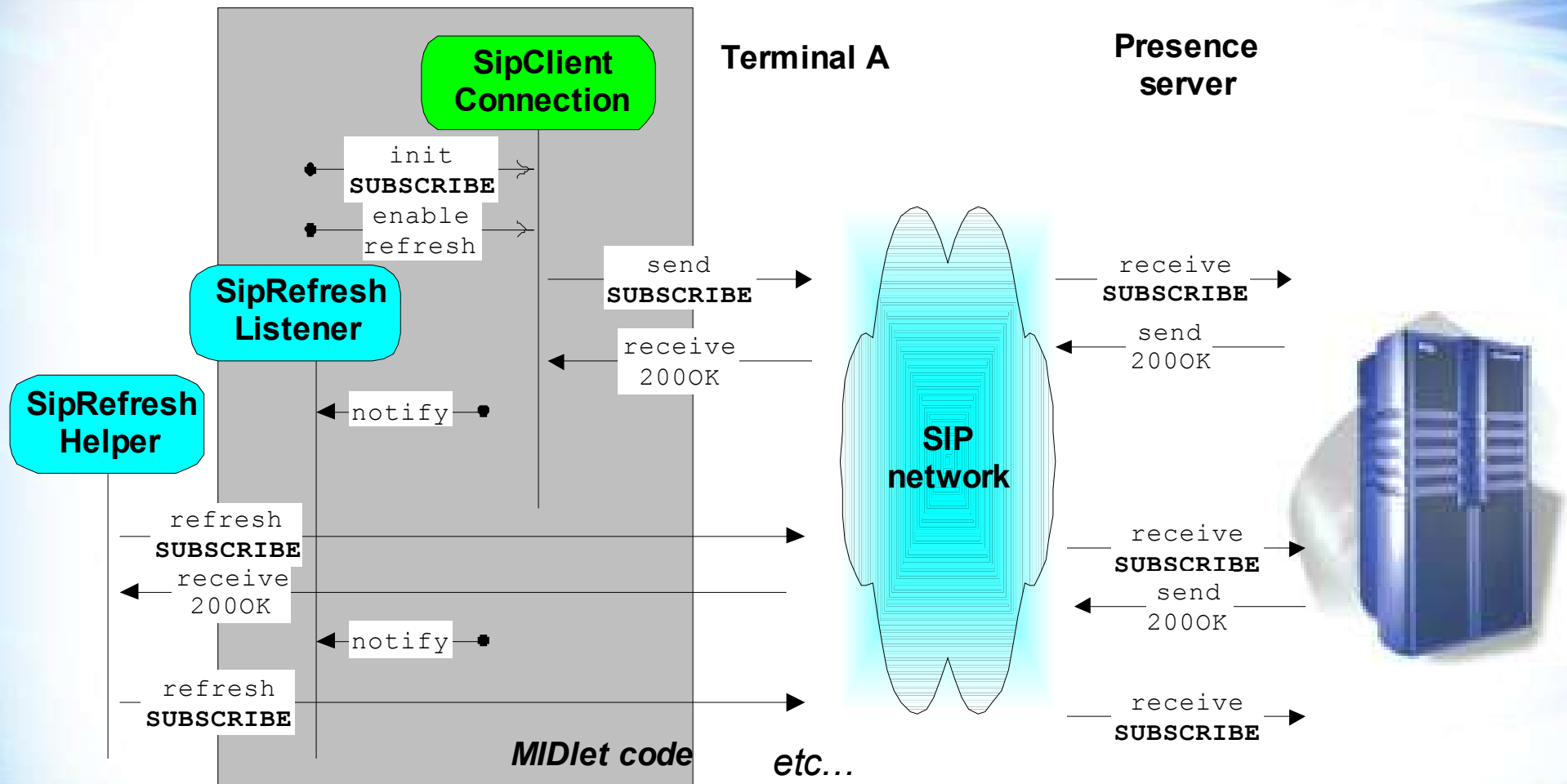
Terminating INVITE



Architecture (Dialog support)

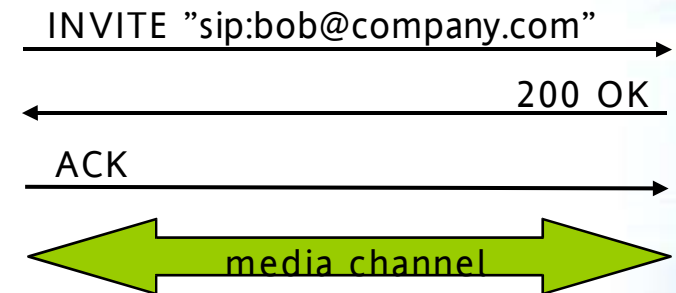


Architecture (Refresh support)



Applications

- Enable innovative SIP application development for operators/3rd parties
- Utilize the Java knowledge on both client and server side (JAIN SIP, SIP Servlets)
- Utilize SIP in peer-to-peer networking as generic rendez-vous and session setup
- Java apps live together with native apps (compare WMA & SMS Messages)
- Core SIP apps usually still in native (e.g. voice/video calls)
- Supporting both native and Java SIP applications not obvious:
 - How to share the SIP resources
 - How to dispatch requests to applications
 - How to share the user's SIP identity and data (phonebook, presence, etc.)



Applications ...

- "Independent" Java SIP applications
 - Chat rooms with anonymous identity. The Java MIDlet registers to a particular discussion forum and is automatically woken up when a new message arrives. This enables chat and group chat applications.
 - Subscribing remote events news/sport/info channels. SIP event notification mechanisms can be used for several types of subscribed information: news flashes, sport event follower, remote alerts, and monitoring
- End-to-end applications
 - Share the system user SIP identity, with all applications
 - Game invitations. The Java game uses SIP to invite another player to the game. A particular game server is not needed, since the SIP network provides the rendezvous mechanism.
 - "Push to talk" type of services, combining SIP and Multimedia services



JSR180 & 3GPP(2) IMS networks

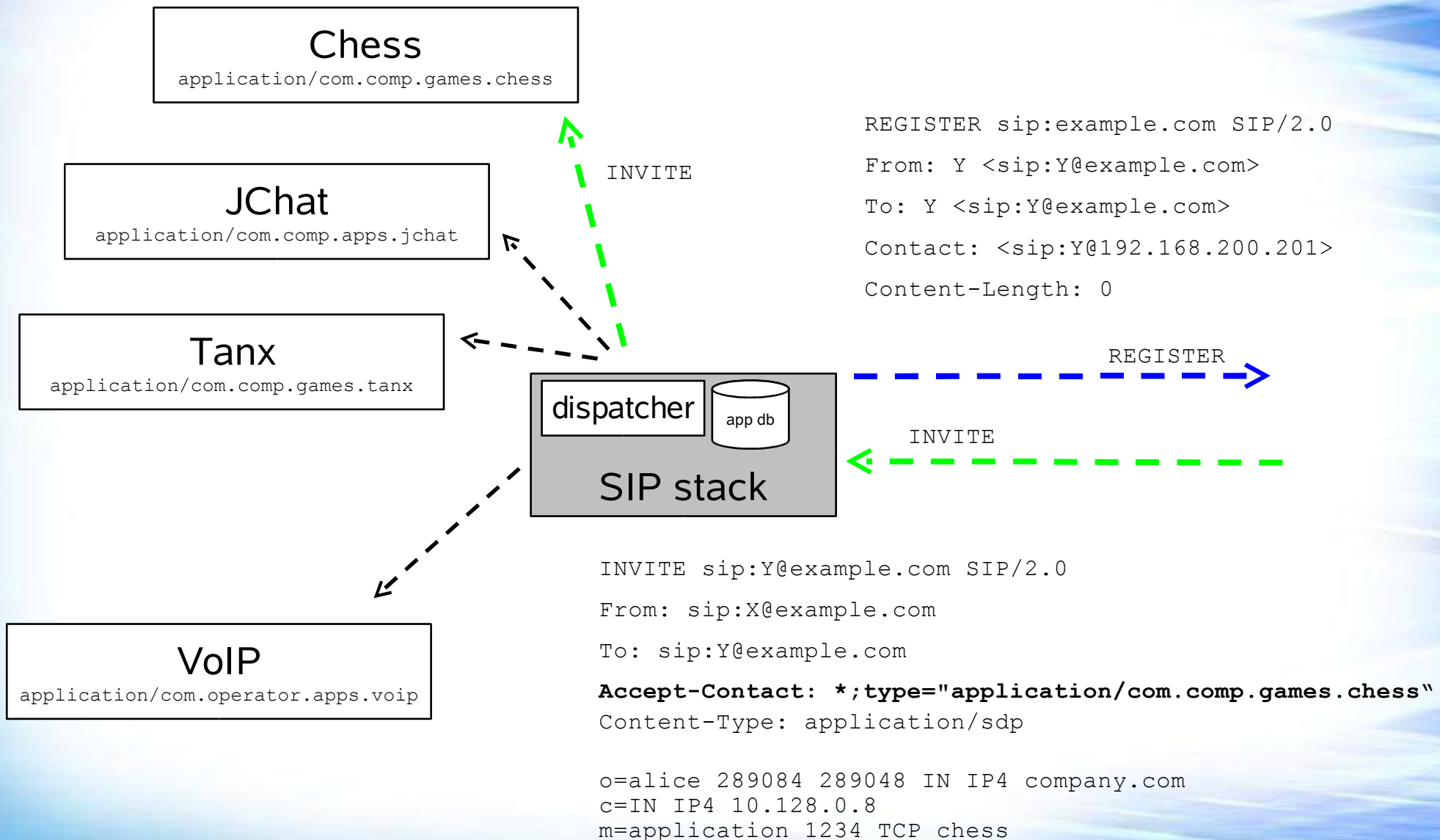
- Assumption: The terminal does the SIP REGISTER (on boot-up) and Java apps share that registration
- Basic SIP signaling support with JSR180 ok: INVITE, 183, 200, PRACK, ACK, UPDATE, ...
- Non real-time services ok, when using one shared "best-effort" PDP context
- Considering real-time services there may be some shortcomings with current APIs:
 - No QoS / resource reservation API (for Secondary PDP context)
 - SDP offer/answer model can be hard to implement
 - Where to get the SDP offer i.e. media capabilities
 - End-to-end streaming not supported

SIP application addressing

- User has one address-of-record `sip:user@operator.com`
- Terminal uses one listening port 5060
- Terminal has several (dynamically installed) applications sharing SIP resources
 - not feasible to REGISTER and open new listening port for every application
 - media type not unique, like SDP, Multipart/*,
 - several apps can utilize e.g. push-to-talk features
- Utilize SIP WG callee capabilities / caller preferences
- Apply same proxy routing mechanism in-terminal
 - Use the Accept-Contact header in caller prefs
- Application registers (app-id) to the terminal SIP stack
 - Routing rules (Accept-Contact or SDP)

...example on next slide

Example (game invite)



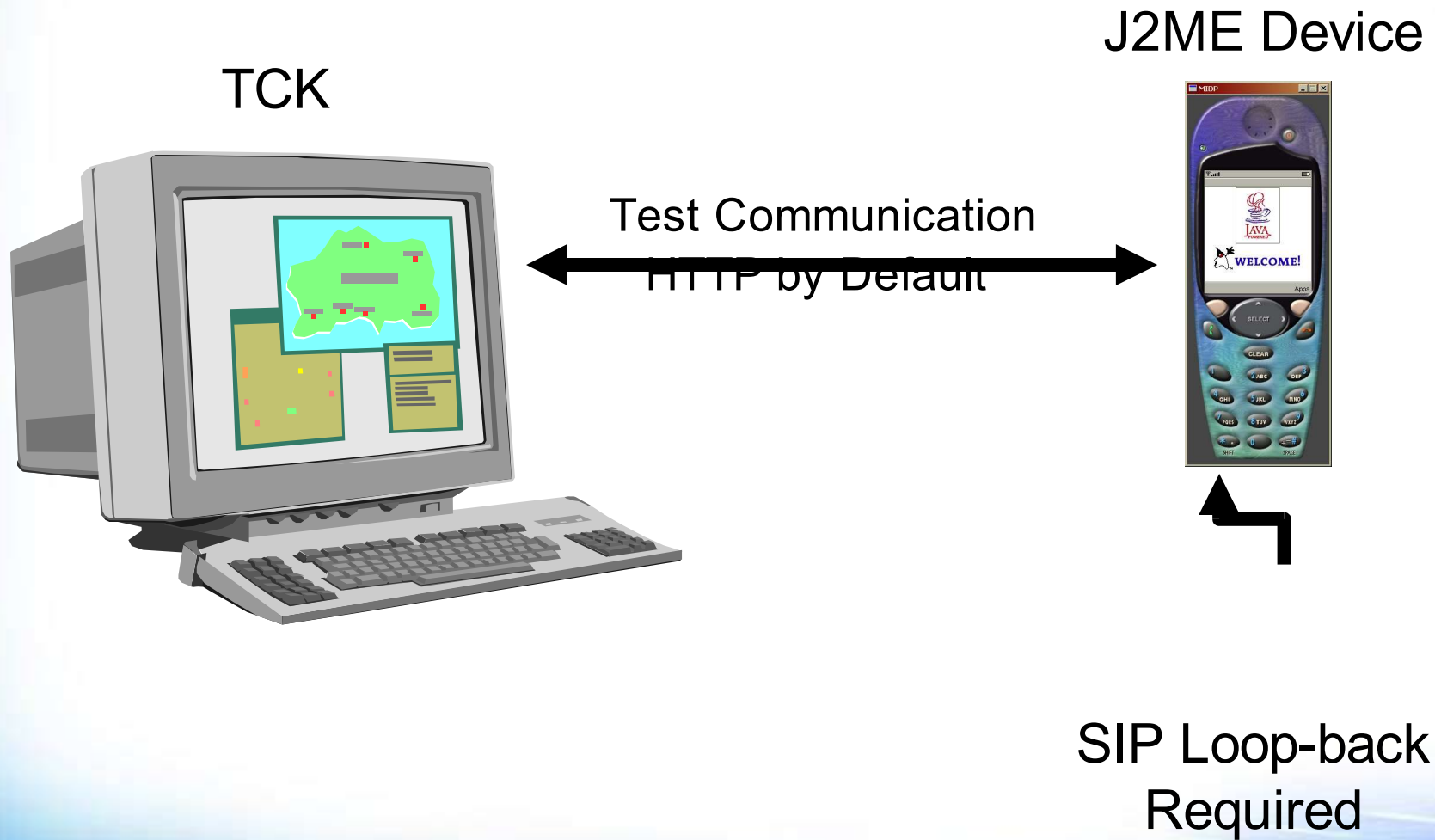
Reference Implementation

- All Java implementation
- Optional Package that can be integrated in standard MIDP2.0/CLDC environment
- Developed on SUN MIDP2.0/CLDC1.0.4 RI
- RI SIP stack implements limited set of RFC3261 functionality
 - essential parts in RFC to act as proof of concept
 - real SIP message transport in IP network
- UDP transport (`UDPDatagramConnection` in MIDP2.0)
- Methods tested so far: REGISTER, INVITE, ACK, BYE, CANCEL, SUBSCRIBE, NOTIFY, MESSAGE, REFER, ...
- Configurable user address, display name, outbound proxy
- Multiple listening points/MIDlet possible (`SipConnectionNotifier`)
- Local-loop support for TCK

Technology Compatibility Kit

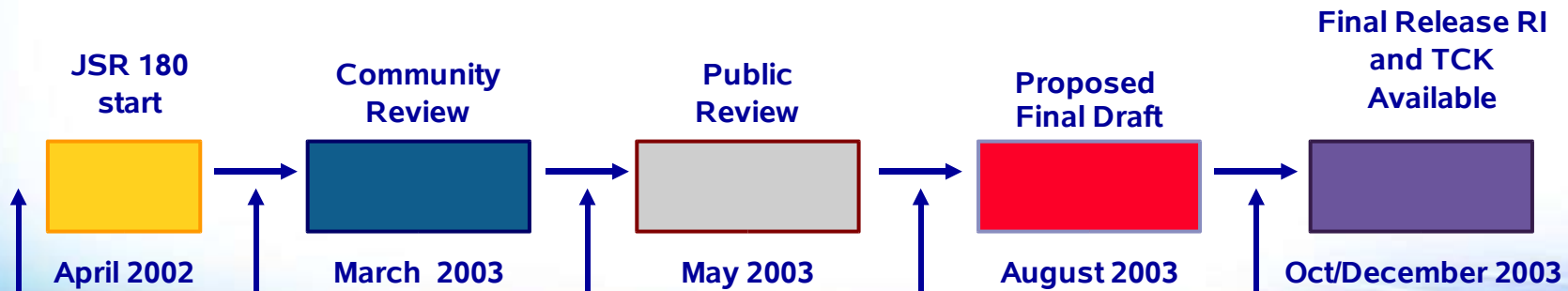
- TCK is designed to test the API not SIP system
- Includes two test modes:
 - MIDP
 - CLDC
- Build on top of Java TestHarness 3.1.2
- TCK contains:
 - API tests
 - Functional tests
- Contains around 300 tests
- Licensing
<http://www.forum.nokia.com/java/jsr180>

Structure of the TCK



JSR180 status

- JSR180 Final Release 10 Dec, 2003
- <http://www.forum.nokia.com/java/jsr180>
- JSR180 is in maintenance phase
 - Follow up feedback from implementers
 - Revise the spec according to needed changes
 - Update RI / TCK accordingly
- All feedback is very welcome:
jsr-180-comments@jcp.org



Thank you!

Q & A